

On the coupled thermo-hydro-mechanical behaviors of layered porous media by the transformed differential quadrature method

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ABSTRACT

In this paper, the thermo-hydro-mechanical (THM) problem of layered porous media is investigated by using the transformed differential quadrature method (TDQM). Starting from the THM governing equations in cylindrical coordinates, the partial differential equations in the transformed domain are obtained by using the Hankel integral transform. Then, the temporal domain and the spatial domain along depth are discretized according to the proposed TDQM, and the partial differential equations in the transformed domain are converted into algebraic equations at sampling points. By introducing load conditions, boundary conditions and continuity conditions, the matrix equation to solve the coupled THM problem of layered porous media is obtained. Finally, case studies are carried out to verify the TDQM solution, and to further discuss the influences of thermal and hydraulic parameters including the linear expansion coefficient of fluid and skeleton, permeable coefficient of fluid and porosity on the THM behaviors of layered porous media.

Introduction

Research on the thermo-hydro-mechanical behaviors of geomaterials has always been a popular and important subject in geotechnical engineering [1–3], and the thermal effect and influence of underground water are considered in various energy geotechnics including energy piles and geothermal power generation. Biot [4] proposed the basic governing equations for the coupled THM problem in porous media in 1956, and provided a clear definition and relatively complete physical explanation of the coupling coefficients. Subsequently, Biot [5] further extended his theory to the coupling problem in poroelastic media containing viscous fluids. The governing equations of Biot's theory is very complex, which are partial differential equations coupled with solids, fluid and temperature changes. It is very difficult to directly solve the governing equations mathematically. Therefore, many scholars have simplified and revised the theories, proposing different sets of thermo-hydro-mechanical coupling governing equations, e.g., Bear & Corapcioglu [6], Palciauskas & Domenico [7], Booker & Savvidou [8], McTigue [9], Kurashige [10], Smith & Booker [11], Jiang & Rajapakse [12], Seneviratne et al. [13], Zhou et al. [14], He & Jin [15], Yang et al. [16], and so on. The above theories are conducted under the framework of Biot's theory and share common assumptions, such as small temperature changes, local thermal equilibrium (LTE), and temperature variation rate being the main factor affecting excess pore fluid pressure. The differences between them are whether considering the compressibility of solid and fluid, heat convection and thermo-osmosis.

Based on the governing equations proposed above, scholars have applied different methods to solve the THM coupling problem in

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Nomenclature

F_0	intensity of concentrated point loads
p_0	intensity of uniformly distributed circular load
Q_0	strength of point heat source
θ_0	circular temperature increment
E	Young's modulus
λ, G	Lamé constants
μ	Poisson's ratio
α_s, α_f	linear expansion coefficient of solid skeleton and fluid
σ	excess pore fluid pressure
K_θ	thermal conductivity coefficient
K_f	coefficient of fluid flow
k_w	permeable coefficient
γ_w	gravity of fluid
ρ	mass density
C	specific heat
r, θ, z	cylindrical coordinates
$\sigma_{ij}(i, j = r, \theta, z)$	stress components
u_r, u_θ, u_z	displacement components
ε_v	volume strain
θ	temperature increment
κ	thermal diffusivity coefficient
t	time
Q_θ	total heat flow in the z-direction
Q_f	total fluid flow in the z-direction
∇^2	Laplace operator under cylindrical coordinates
n	porosity
$g(r, z, t), \bar{g}^m(\xi, z, t)$	function and corresponding function in the Hankel transformed domain
ξ	Hankel transform parameter with respect to r
$J_m(\cdot)$	the m th order Bessel function
$\bar{u}_r, \bar{u}_z, \bar{\theta}$	displacement components in the Hankel transformed domain
L_n	number of layers
N_z	number of sampling points along depth
N_t	number of sampling points in a time step
$\bar{F} _{i,j,n}$	function $\bar{F}$ at the i th sampling point along depth in the n th layer, and the time is at the j th point
A_{ik}, A_{ik}^2	weighting coefficients of the first-order and second-order for depth
B_{jk}, B_{jk}^2	weighting coefficients of the first-order and second-order for time
$\mathbf{W}_n$	the variable vector of the n th layer to be sought
$\mathbf{U}_{nr}, \mathbf{U}_{nz}, \boldsymbol{\theta}_n, \boldsymbol{\sigma}_n$	vectors of the n th layer for radial displacements, vertical displacements, temperatures and excess pore fluid pressures
$\hat{\theta}_n, \hat{\mathbf{U}}_{nr}, \hat{\mathbf{U}}_{nz}$	initial temperature vector, initial radial displacement vector, initial vertical displacement vector
$\mathbf{A}, \mathbf{A}^2, \hat{\mathbf{A}}, \mathbf{A}^2, \mathbf{B}_{ij}, \hat{\mathbf{B}}, \hat{\mathbf{B}}, \hat{\mathbf{C}}, \hat{\mathbf{C}}$	weighting coefficient matrices
$\mathbf{K}_n$	stiffness matrix of the n th layer
$\mathbf{F}_n$	load vector of the n th layer
r_0	the radius of the loading
R_o	the distance of from the observation point to the point heat source
P_k	sensitivity evaluation criterion for the k -th parameter
R	output result
$\mathbf{K}_g$	global stiffness matrix
$\mathbf{F}_g$	global load vector
$H^{-1}(\cdot)$	the Hankel inversion
N	input parameter for computation
$\xi_i(i = 0 \sim N)$	the i th zero points of the m th Bessel function
ω_j, ϖ_j	the j th Gauss weight coefficients and Gauss points
K	Kelvins for temperature
u_z^*, τ, θ^*	normalized displacement, normalized time and normalized temperature

porous media. Bear & Corapcioglu [6] established a mathematical model to study the excess pore fluid pressure, surface settlements, horizontal displacements, and temperatures when high-temperature fluids are injected into isotropic poroelastic media. Booker & Savvidou [8] took use of the integral transformation to deduce an analytical solution for isotropic porous media under a point heat source, and further obtained an approximate solution for a cylindrical heat source by integrating point heat sources within a certain volume. Considering the compressibility of solid and fluid, McTigue [9] solved the THM coupling problem of poroelastic media with a heat source and a constant heat flow acting on the surface. Kurashige [10] established a thermoelastic coupling theory, and further studied the THM coupling problem of injecting a heat flow into a saturated half-space through a spherical cavity. Smith & Booker [11] derived the Green function for thermal consolidation problems in homogeneous, isotropic, porous media using Laplace integral transform. Jiang & Rajapakse [12], Seneviratne et al. [13], Wang & Papamichos [17], Rihbinder [18], Blond et al. [19] also obtained similar analytical solutions for homogeneous, isotropic, thermoelastic, porous media under different boundary conditions. Lee & Mei [20,21] obtained an analytical solution for the linear THM coupling problem in porous media based on the homogenization theory. Zhang et al. [22] conducted an elastic-plastic analysis of the one-dimensional thermal consolidation and discussed key influencing parameters. Selvadurai & Suvorov [1] conducted elastic and elastic-plastic studies on the THM response of porous media under different conditions. Bai & Abousseiman [23] provided analytical solutions for three cases of THM problems under one-dimensional conditions, and explored the differences and applicability of fully-coupled, semi-coupled, and non-coupled theories. Zhou et al. [24] considered the influence of thermo-osmosis and provided temperature and stress solutions for media with a cylindrical or a spherical cavity. Bai [25] derived a solution of saturated media under cyclic thermal loadings using Laplace transform and Gauss-Legendre integration. El-Zein [26] applied the boundary element method (BEM) to obtain an analytical method for solving the fully-coupled equations of thermoelastic consolidation of layered media. Considering thermo-osmosis and mechano-caloric couplings, Song et al. [2,3] obtained the axisymmetric THM solution when point and circular heat sources acting on the interior of porous media.

There have been many achievements in the study of THM in porous media. However, existing studies often consider the form of thermal load as a stable or decaying heat source, and there is little research on the behavior of the media under temperature loads. Therefore, this article proposed a novel TDQM solution for the coupled THM problem in layered porous media under temperature loads. The TDQM needs less algebraic equations to obtain the accurate results compared with the traditional differential quadrature method (DQM), so that the computational efficiency and accuracy are greatly improved. After verify the TDQM solution, the sensitivity analyses are carried out to further investigate the influences of the linear expansion coefficients, permeable coefficient of fluid and porosity on the coupled THM behaviors of porous media.

Solution process

Problem description

The layered isotropic porous media filled with fluid are shown in Fig. 1. The displacement of the bottom is zero, and there is no heat transfer on the bottom. Time-varying mechanical loads and heat sources or temperature loads are applied on its surface or interior. In this paper, various load types including concentrated point loads F_0 , uniformly distributed circular loads p_0 , point heat source Q_0 , circular temperature increments θ_0 are considered. Heat sources can be constant or decaying with time. Consolidation and heat transfer occur under the external loads, and coupled thermo-hydro-mechanical behaviors of the media are investigated herein.

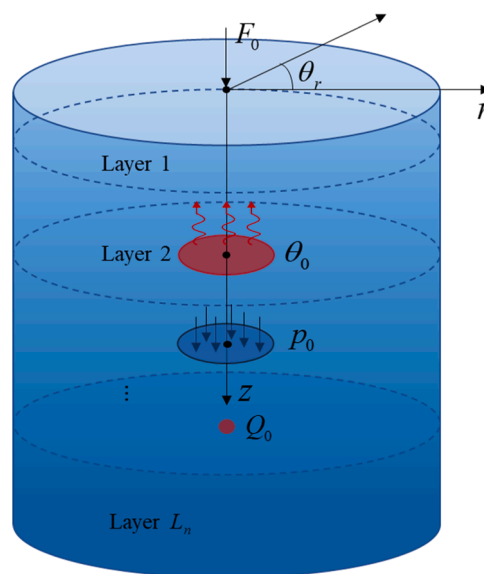


Fig. 1. Multilayered media under thermal loadings and mechanical loadings.

Transformed differential quadrature solution

The equilibrium equations without body forces are

$$\frac{\partial \sigma_r}{\partial r} + \frac{\partial \sigma_{rz}}{\partial z} + \frac{\sigma_r - \sigma_{\theta_r}}{r} = 0 \quad (1a)$$

$$\frac{\partial \sigma_z}{\partial z} + \frac{\partial \sigma_{rz}}{\partial r} + \frac{\sigma_{rz}}{r} = 0 \quad (1b)$$

in which $\sigma_{ij}(i, j = r, \theta, z)$ are stress components in cylindrical coordinates.

Considering the effective stresses principle and the thermal effect, the stress components in porous media can be expressed by the displacements and the temperature as the following formulae,

$$\sigma_r = \lambda \varepsilon_v + 2G \frac{\partial u_r}{\partial r} - \beta \theta - \sigma \quad (2a)$$

$$\sigma_{\theta} = \lambda \varepsilon_v + 2G \frac{u_r}{r} - \beta \theta - \sigma \quad (2b)$$

$$\sigma_z = \lambda \varepsilon_v + 2G \frac{\partial u_z}{\partial z} - \beta \theta - \sigma \quad (2c)$$

$$\sigma_{rz} = G \left(\frac{\partial u_r}{\partial z} + \frac{\partial u_z}{\partial r} \right) \quad (2d)$$

in which $\lambda = E\mu / (1 + \mu)(1 - 2\mu)$ and $G = E / 2(1 + \mu)$ are Lamé constants; $\varepsilon_v = \frac{\partial u_r}{\partial r} + \frac{u_r}{r} + \frac{\partial u_z}{\partial z}$ is the volume strain; $\beta = 2G\alpha_s(1 + \mu) / (1 - 2\mu)$; θ is the temperature increment; σ is excess pore fluid pressure; and α_s is the linear expansion coefficient of the solid skeleton;

It is noted that Eq. (2a–2d) can be generated to the Hooke's law for isotropic media without considering the temperature increments and excess pore fluid pressure. The heat flow in the z -direction at any time can be expressed:

$$Q_{\theta} = -K_{\theta} \int_0^t \frac{\partial \theta}{\partial z} dt \quad (3)$$

where Q_{θ} is the heat flow, K_{θ} is the coefficient of heat conduction.

Similarly, based on the Darcy's law of fluid flow, the fluid flow in the z -direction is defined as

$$Q_f = -K_f \int_0^t \frac{\partial \sigma}{\partial z} dt \quad (4)$$

where Q_f is the total fluid flow; $K_f = k_w / \gamma_w$ is the coefficient of fluid flow, in which k_w is the permeable coefficient, and γ_w is the gravity of fluid.

The heat diffusion in media can be expressed as

$$\frac{\partial \theta}{\partial t} = \kappa \left(\frac{\partial^2 \theta}{\partial r^2} + \frac{1}{r} \frac{\partial \theta}{\partial r} + \frac{\partial^2 \theta}{\partial z^2} \right) \quad (5)$$

where $\kappa = K_{\theta} / \rho C$ represents the thermal diffusivity coefficient, in which C is the specific heat and ρ is mass density.

In this study, the solid skeleton and pore fluid are seen as incompressible, that is, the volume of solid and fluid do not change under external mechanical loads. By contrast, the temperature increment leads to the volume expansion or contraction of the media. In this way, the seepage equation considering temperature changes is

$$\frac{\partial \varepsilon_v}{\partial t} + \alpha_u \frac{\partial \theta}{\partial t} = \frac{k_w}{\gamma_w} \nabla^2 \sigma \quad (6)$$

where $\alpha_u = 3\alpha_s(n - 1) - 3\alpha_f n$, in which α_f is the linear expansion coefficient of the fluid, and n is the porosity.

Without considering the heat diffusion, one has $\alpha_u = 0$ and Eq. (6) is transformed into seepage equation in Biot's theory of consolidation. By substituting Eqs. (2a–2d) in Eqs. (1a,1b), we have

$$\nabla^2 u_r + \frac{1}{1 - 2\mu} \frac{\partial \varepsilon_v}{\partial r} - \frac{1}{r^2} u_r - \frac{\beta}{G} \frac{\partial \theta}{\partial r} = 0 \quad (7a)$$

$$\nabla^2 u_z + \frac{1}{1 - 2\mu} \frac{\partial \varepsilon_v}{\partial z} - \frac{\beta}{G} \frac{\partial \theta}{\partial z} = 0 \quad (7b)$$

where $\nabla^2 = \frac{\partial^2}{\partial r^2} + \frac{1}{r} \frac{\partial}{\partial r} + \frac{\partial^2}{\partial z^2}$ is the Laplace operator.

Eqs. (5-7) are the governing equations of the coupled THM problem of porous media. To solve the governing equations, we apply the TDQM which greatly reduce the number of algebraic equations compared with the traditional DQM. Based on the theory of the TDQM, the partial derivatives with respect to r are firstly dealt with. The definitions of the Hankel transform and its inversion are

$$\bar{g}^m(\xi, z, t) = \int_0^\infty g(r, z, t) J_m(\xi r) r dr \quad (8a)$$

$$g(r, z, t) = \int_0^\infty \bar{g}^m(\xi, z, t) J_m(\xi r) \xi d\xi \quad (8b)$$

where $g(r, z, t)$ is a function in physical domain, and $\bar{g}^m(\xi, z, t)$ is the transformed function; ξ is the Hankel transform parameter and $J_m(\cdot)$ is the m th order Bessel function.

Then we have the transformed governing equations which are dependent on t and z as follows,

$$\frac{\partial^2 \bar{u}_r^1}{\partial z^2} - \frac{\lambda + 2G}{G} \xi^2 \bar{u}_r^1 - \frac{\lambda + G}{G} \xi \frac{\partial \bar{u}_z^0}{\partial z} + \frac{\beta}{G} \xi \bar{\theta}^0 + \frac{1}{G} \xi \bar{\sigma}^0 = 0 \quad (9a)$$

$$\frac{\lambda + G}{G} \xi \frac{\partial \bar{u}_r^1}{\partial z} + \frac{\lambda + 2G}{G} \frac{\partial^2 \bar{u}_z^0}{\partial z^2} - \xi^2 \bar{u}_z^0 - \frac{\beta}{G} \frac{\partial \bar{\theta}^0}{\partial z} - \frac{1}{G} \frac{\partial \bar{\sigma}^0}{\partial z} = 0 \quad (9b)$$

$$\kappa \frac{\partial^2 \bar{\theta}^0}{\partial z^2} - \kappa \xi^2 \bar{\theta}^0 - \frac{\partial \bar{\theta}^0}{\partial t} = 0 \quad (9c)$$

$$\xi \frac{\partial \bar{u}_r^1}{\partial t} + \frac{\partial^2 \bar{u}_z^0}{\partial z \partial t} + \alpha_u \frac{\partial \bar{\theta}^0}{\partial t} - \frac{k_w}{\gamma_w} \frac{\partial^2 \bar{\sigma}^0}{\partial z^2} + \frac{k_w}{\gamma_w} \xi^2 \bar{\sigma}^0 = 0 \quad (9d)$$

where 0 and 1 represent the order of the Bessel function in the Hankel transform. $\bar{u}_r^1$, $\bar{u}_z^0$, $\bar{\theta}^0$ and $\bar{\sigma}^0$ are written as $\bar{u}_r$, $\bar{u}_z$, $\bar{\theta}$ and $\bar{\sigma}$ for brevity, respectively.

It is noted that the dimension of the governing equations has been reduced in the transformed domain, that is, the spatial coordinate only contains depth z . As a result, the equations can be conveniently discretized only along depth in spatial domain. In addition, we discretize the temporal domain according to the differential quadrature rule [27], which is the same with the procession of the incremental differential quadrature method (IDQM). This procession is quite suitable for solving time-dependent PDEs, especially for the problems with large time duration [28–30] and problems with harmonic loadings [31]. The temporal domain is divided into several time step and then every step is discretized according to the differential quadrature rule for a better convergence and accuracy. By contrast, if one process the temporal domain with the Laplace transform, the results obtained by different numerical transform inversion will be quite different [27]. The TDQM method and the Laplace method can obtain satisfactory results when the medium is under step or decaying thermal loadings. However, for time-harmonic temperature loads, the convergence and the computational efficiency of the TDQM are superior to the Laplace approach [31]. In this study, the scheme #2 of the TDQM in Ref. [27] (only applying the Fourier transform or Hankel transform to PDEs to deal with the spatial derivatives) is applied since it could be applicable for more general thermal loadings. The number of sampling points of every step for time t is 3, and that for depth z is 14, which are determined by a convergence trial.

Subsequently, the transformed partial differential equations are discretized according to the discretization rule of the TDQM. Then, at any sampling point in the n th layer, we have

$$\sum_{k=1}^{N_z} A_{ik}^2 \bar{u}_r|_{k,j,n} - \frac{\lambda_n + 2G_n}{G_n} \xi^2 \bar{u}_r|_{i,j,n} - \quad (10a)$$

$$\frac{\lambda_n + G_n}{G_n} \xi \sum_{k=1}^{N_z} A_{ik} \bar{u}_z|_{k,j,n} + \frac{\beta_n}{G_n} \xi \bar{\theta}|_{i,j,n} + \frac{1}{G_n} \xi \bar{\sigma}|_{i,j,n} = 0$$

$$\frac{\lambda_n + G_n}{G_n} \xi \sum_{k=1}^{N_z} A_{ik} \bar{u}_r|_{k,j,n} + \frac{\lambda_n + 2G_n}{G_n} \sum_{k=1}^{N_z} A_{ik}^2 \bar{u}_z|_{k,j,n} - \quad (10b)$$

$$\xi^2 \bar{u}_z|_{i,j,n} - \frac{\beta_n}{G_n} \sum_{k=1}^{N_z} A_{ik} \bar{\theta}|_{k,j,n} - \frac{1}{G_n} \sum_{k=1}^{N_z} A_{ik} \bar{\sigma}|_{k,j,n} = 0$$

$$\kappa_n \sum_{k=1}^{N_z} A_{ik}^2 \bar{\theta}|_{k,j,n} - \kappa_n \xi^2 \bar{\theta}|_{i,j,n} - \left(B_{j1} \bar{\theta}|_{i,1,n} + \sum_{k=2}^{N_t} B_{jk} \bar{\theta}|_{i,k,n} \right) = 0 \quad (10c)$$

$$\xi \left(B_{j1} \bar{u}_r|_{i,1,n} + \sum_{k=2}^{N_t} B_{jk} \bar{u}_r|_{i,k,n} \right) + B_{j1} \sum_{k=1}^{N_z} A_{ik} \bar{u}_z|_{k,1,n} + \sum_{k=2}^{N_t} B_{jk} \sum_{l=1}^{N_z} A_{il} \bar{u}_z|_{l,k,n} + \alpha_{un} \left(B_{j1} \bar{\theta}|_{i,1,n} + \sum_{k=2}^{N_t} B_{jk} \bar{\theta}|_{i,k,n} \right) - \frac{k_{wn}}{\gamma_{wn}} \sum_{k=1}^{N_z} A_{ik}^2 \bar{\sigma}|_{k,j,n} + \frac{k_{wn}}{\gamma_{wn}} \xi^2 \bar{\sigma}|_{i,j,n} = 0 \quad (10d)$$

where N_z and N_t are the number of sampling points along z for the n th layer and the number of points of a time step, respectively; $(\cdot)|_{i,j,n}$ denotes the variables of the i th point along z , the j th point along time and inside the n th layer; the subscript n indicates that the parameters belong to the n th layer; A_{ik} , A_{ik}^2 , B_{jk} and B_{jk}^2 are weighting coefficients, which could be obtained by the formulae in Zhao & Ai [27].

We can rewrite Eq. (10a–10d) as

$$\left[\hat{\mathbf{A}}^2 - \frac{\lambda_n + 2G_n}{G_n} \xi^2 \mathbf{I} \quad -\frac{\lambda_n + G_n}{G_n} \xi \hat{\mathbf{A}} \quad \frac{\beta_n}{G_n} \xi \mathbf{I} \quad \frac{1}{G_n} \xi \mathbf{I} \right] \mathbf{W}_n = \mathbf{0} \quad (11a)$$

$$\left[\frac{\lambda_n + G_n}{G_n} \xi \hat{\mathbf{A}} \quad \frac{\lambda_n + 2G_n}{G_n} \hat{\mathbf{A}}^2 - \xi^2 \mathbf{I} \quad -\frac{\beta_n}{G_n} \hat{\mathbf{A}} \quad -\frac{1}{G_n} \hat{\mathbf{A}} \right] \mathbf{W}_n = \mathbf{0} \quad (11b)$$

$$\left[\mathbf{0} \quad \mathbf{0} \quad \kappa_n \hat{\mathbf{A}}^2 - \kappa_n \xi^2 \mathbf{I} - \hat{\mathbf{B}} \quad \mathbf{0} \right] \mathbf{W}_n = \hat{\mathbf{B}} \hat{\theta}_n, \quad (11c)$$

$$\left[\xi \hat{\mathbf{B}} \quad \hat{\mathbf{C}} \quad \alpha_{un} \hat{\mathbf{B}} \quad -\frac{k_{wn}}{\gamma_{wn}} \hat{\mathbf{A}}^2 + \frac{k_{wn}}{\gamma_{wn}} \xi^2 \mathbf{I} \right] \mathbf{W}_n = -\xi \hat{\mathbf{B}} \hat{\mathbf{U}}_{nr} - \hat{\mathbf{C}} \hat{\mathbf{U}}_{nz} - \alpha_{un} \hat{\mathbf{B}} \hat{\theta}_n \quad (11d)$$

where $\mathbf{W}_n = [\mathbf{U}_{nr} \quad \mathbf{U}_{nz} \quad \boldsymbol{\theta}_n \quad \boldsymbol{\sigma}_n]^T$ is the variables vector, and $\mathbf{U}_{nr}$, $\mathbf{U}_{nz}$, $\boldsymbol{\theta}_n$ and $\boldsymbol{\sigma}_n$ are vectors composed of radial displacements, vertical displacements, temperatures and excess pore fluid pressure, respectively; $\hat{\theta}_n = [\bar{\theta}|_{1,1,n}, \bar{\theta}|_{2,1,n}, \dots, \bar{\theta}|_{N_z,1,n}]^T$ is the initial temperature vector. The coefficients matrices are shown in Appendix A, which are composed of weighting coefficients and dependent on the number of sampling points and depth z .

For the n th layer, the matrix equation can be written as follow by assembling Eq. (11a–11d),

$$\mathbf{K}_n \mathbf{W}_n = \mathbf{F}_n, \quad (12)$$

in which,

$$\mathbf{K}_n = \begin{bmatrix} \hat{\mathbf{A}}^2 - \frac{\lambda_n + 2G_n}{G_n} \xi^2 \mathbf{I} & -\frac{\lambda_n + G_n}{G_n} \xi \hat{\mathbf{A}} & \frac{\beta_n}{G_n} \xi \mathbf{I} & \frac{1}{G_n} \xi \mathbf{I} \\ \frac{\lambda_n + G_n}{G_n} \xi \hat{\mathbf{A}} & \frac{\lambda_n + 2G_n}{G_n} \hat{\mathbf{A}}^2 - \xi^2 \mathbf{I} & -\frac{\beta_n}{G_n} \hat{\mathbf{A}} & -\frac{1}{G_n} \hat{\mathbf{A}} \\ \mathbf{0} & \mathbf{0} & \kappa_n \hat{\mathbf{A}}^2 - \kappa_n \xi^2 \mathbf{I} - \hat{\mathbf{B}} & \mathbf{0} \\ \xi \hat{\mathbf{B}} & \hat{\mathbf{C}} & \alpha_{un} \hat{\mathbf{B}} & -\frac{k_{wn}}{\gamma_{wn}} (\hat{\mathbf{A}}^2 - \xi^2 \mathbf{I}) \end{bmatrix} \quad (13)$$

$$\mathbf{F}_n = [\mathbf{0} \quad \mathbf{0} \quad \hat{\mathbf{B}} \hat{\theta}_n \quad -\xi \hat{\mathbf{B}} \hat{\mathbf{U}}_{nr} - \hat{\mathbf{C}} \hat{\mathbf{U}}_{nz} - \alpha_{un} \hat{\mathbf{B}} \hat{\theta}_n]^T. \quad (14)$$

Then, considering the saturated media have L_n layers, we can obtain the solving equation of the media,

$$\begin{bmatrix} \mathbf{K}_1 & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \mathbf{K}_2 & & \vdots \\ \vdots & & \ddots & \\ \mathbf{0} & \cdots & & \mathbf{K}_{L_n} \end{bmatrix} \begin{bmatrix} \mathbf{W}_1 \\ \mathbf{W}_2 \\ \vdots \\ \mathbf{W}_{L_n} \end{bmatrix} = \begin{bmatrix} \mathbf{F}_1 \\ \mathbf{F}_2 \\ \vdots \\ \mathbf{F}_{L_n} \end{bmatrix} \quad (15)$$

After introducing specific boundary conditions and initial conditions of the problem, the general variables including displacements and temperatures inside the layered porous media can be obtained. The initial conditions of the problem are

$$u_r|_{t=0} = 0, u_z|_{t=0} = 0, \sigma|_{t=0} = 0, \theta|_{t=0} = 0 \quad (16)$$

The top boundary conditions are

$$\sigma_z|_{1,j,1} = 0, \sigma_{rz}|_{1,j,1} = 0, \sigma|_{1,j,1} = 0, \theta|_{1,j,1} = 0, j = 2 \sim N_t \quad (17)$$

The bottom is impermeable for heat flux and fluid flux, and is fixed for displacement, which means

$$u_z|_{N_z,j,L_n} = 0, u_r|_{N_z,j,L_n} = 0, \frac{\partial \sigma}{\partial z}|_{N_z,j,L_n} = 0, \frac{\partial \theta}{\partial z}|_{N_z,j,L_n} = 0, j = 2 \sim N_t \quad (18)$$

To introduce the stress conditions, we take the integral transform to Eq. (2c,2d):

$$\bar{\sigma}_z = \lambda \xi \bar{u}_r + (\lambda + 2G) \frac{\partial \bar{u}_z}{\partial z} - \beta \bar{\theta} - \bar{\sigma} \quad (19a)$$

$$\bar{\sigma}_{zr} = G \frac{\partial \bar{u}_r}{\partial z} - G \xi \bar{u}_z \quad (19b)$$

Based on Eqs. (17, 19a, 19b), the boundary conditions can be expressed as

$$\left\{ \begin{array}{l} \lambda_1 \xi \bar{u}_r|_{1,j,1} + (\lambda_1 + 2G_1) \sum_{k=1}^{N_z} A_{1k} \bar{u}_z|_{k,j,1} - \beta_1 \bar{\theta}|_{1,j,1} - \bar{\sigma}|_{1,j,1} = 0, \\ -\xi G_1 \bar{u}_z|_{1,j,1} + G_1 \sum_{k=1}^{N_z} A_{1k} \bar{u}_r|_{k,j,1} = 0, \\ \bar{\theta}|_{1,j,1} = 0, \\ \bar{\sigma}|_{1,j,1} = 0, \\ \bar{u}_z|_{N_z,j,L_n} = 0, \\ \bar{u}_r|_{N_z,j,L_n} = 0, \\ \sum_{k=1}^{N_z} A_{N_z,k} \bar{\theta}|_{k,j,L_n} = 0, \\ \sum_{k=1}^{N_z} A_{N_z,k} \bar{\sigma}|_{k,j,L_n} = 0. \end{array} \right. \quad (20)$$

Since the porous elastic media have several layers with different properties, the internal continuity conditions should also be introduced. Adjacent layers (n and $n + 1$ layers) have the following relationships:

$$\left\{ \begin{array}{l} \bar{u}_r|_{N_z,j,n} - \bar{u}_r|_{1,j,n+1} = 0, \\ \bar{u}_z|_{N_z,j,n} - \bar{u}_z|_{1,j,n+1} = 0, \\ \bar{\theta}|_{N_z,j,n} - \bar{\theta}|_{1,j,n+1} = 0, \\ \bar{\sigma}|_{N_z,j,n} - \bar{\sigma}|_{1,j,n+1} = 0, \\ \bar{\sigma}_z|_{N_z,j,n} - \bar{\sigma}_z|_{1,j,n+1} = 0, \\ \bar{\sigma}_{rz}|_{N_z,j,n} - \bar{\sigma}_{rz}|_{1,j,n+1} = 0, \\ \bar{Q}_\theta|_{N_z,j,n} - \bar{Q}_\theta|_{1,j,n+1} = 0, \\ \bar{Q}_f|_{N_z,j,n} - \bar{Q}_f|_{1,j,n+1} = 0. \end{array} \right. \quad (21)$$

By combining the Hankel transform and the stress-strain relationship of the media, Eq. (21) can be transformed into

$$\left\{ \begin{array}{l} \bar{u}_r|_{N_z,j,n} - \bar{u}_r|_{1,j,n+1} = 0, \\ \bar{u}_z|_{N_z,j,n} - \bar{u}_z|_{1,j,n+1} = 0, \\ \bar{\theta}|_{N_z,j,n} - \bar{\theta}|_{1,j,n+1} = 0, \\ \bar{\sigma}|_{N_z,j,n} - \bar{\sigma}|_{1,j,n+1} = 0, \\ \lambda_n \xi \bar{u}_r|_{N_z,j,n} + (\lambda_n + 2G_n) \sum_{k=1}^{N_z} A_{N_z,k} \bar{u}_z|_{k,j,n} - \beta_n \bar{\theta}|_{N_z,j,n} - \bar{\sigma}|_{N_z,j,n} - \\ \lambda_{n+1} \xi \bar{u}_r|_{1,j,n+1} - (\lambda_{n+1} + 2G_{n+1}) \sum_{k=1}^{N_z} A_{1k} \bar{u}_z|_{k,j,n+1} + \beta_{n+1} \bar{\theta}|_{1,j,n+1} + \bar{\sigma}|_{N_z,j,n+1} = 0, \\ -\xi G_n \bar{u}_z|_{N_z,j,n} + G_n \sum_{k=1}^{N_z} A_{N_z,k} \bar{u}_r|_{k,j,n} + \\ \xi G_{n+1} \bar{u}_z|_{1,j,n+1} - G_{n+1} \sum_{k=1}^{N_z} A_{1k} \bar{u}_r|_{k,j,n+1} = 0, \\ -K_{\theta,n} \sum_{k=1}^{N_z} A_{N_z,k} \bar{\theta}|_{k,j,n} + K_{\theta,n+1} \sum_{k=1}^{N_z} A_{1k} \bar{\theta}|_{k,j,n+1} = 0, \\ \frac{k_{w,n}}{\gamma_{w,n}} \sum_{k=1}^{N_z} A_{N_z,k} \bar{\sigma}|_{k,j,n} - \frac{k_{w,n+1}}{\gamma_{w,n+1}} \sum_{k=1}^{N_z} A_{1k} \bar{\sigma}|_{k,j,n+1} = 0. \end{array} \right. \quad (22)$$

Then the mechanical load conditions and the temperature conditions are considered. Several typical load forms and their forms in the transformed domain can be seen in Tables 1–3 of the study by Zhao & Ai [32]. The equations of internal layers are replaced to introduce the loadings. For example, if there is a uniform circular mechanical load between the L_p th and the $(L_p + 1)$ th layers, we have

Table 1

The parameters of the stratified layers.

Layer	E/MPa	μ	$K_0/W \cdot \text{m}^{-1} \cdot \text{K}^{-1}$	depth/m
1	10	0.30	1	2
2	20	0.25	2	4
3	35	0.20	2	4

$$\begin{aligned} & \lambda_{L_p} \xi \bar{u}_r|_{N_z, j, L_p} + (\lambda_{L_p} + 2G_{L_p}) \sum_{k=1}^{N_z} A_{N_z, k} \bar{u}_z|_{k, j, L_p} - \beta_{L_p} \bar{\theta}|_{N_z, j, L_p} - \bar{\sigma}|_{N_z, j, L_p} - \lambda_{L_p+1} \xi \bar{u}_r|_{1, j, L_p+1} \\ & - (\lambda_{L_p+1} + 2G_{L_p+1}) \sum_{k=1}^{N_z} A_{1, k} \bar{u}_z|_{k, j, L_p+1} + \beta_{L_p+1} \bar{\theta}|_{1, j, L_p+1} + \bar{\sigma}|_{N_z, j, L_p+1} = -\frac{p_0 a J_1(\xi a)}{2\pi \xi} \end{aligned} \quad (23)$$

Subsequently, the global matrix equation is obtained if the boundary conditions and the load conditions are introduced, which is written as

$$\mathbf{K}_g \mathbf{W}_g = \mathbf{F}_g, \quad (24)$$

where $\mathbf{K}_g$ is the stiffness matrix and $\mathbf{F}_g$ is the global load vector.

Obviously, $\mathbf{W}_g$ is

$$\mathbf{W}_g = \begin{bmatrix} \bar{u}_r|_{1,2,1}, \bar{u}_r|_{2,2,1}, \dots, \bar{u}_r|_{N_z,2,1}, \bar{u}_r|_{1,3,1}, \dots, \bar{u}_r|_{N_z,N_f,1}, \bar{u}_z|_{1,2,1}, \dots, \bar{u}_z|_{N_z,N_f,1}, \bar{\theta}|_{1,2,1}, \dots, \bar{\theta}|_{N_z,N_f,1}, \dots, \bar{\sigma}|_{N_z,N_f,1} \\ \bar{u}_r|_{1,2,2}, \bar{u}_r|_{2,2,2}, \dots, \bar{u}_r|_{N_z,2,2}, \bar{u}_r|_{1,3,2}, \dots, \bar{u}_r|_{N_z,N_f,2}, \bar{u}_z|_{1,2,2}, \dots, \bar{u}_z|_{N_z,N_f,2}, \bar{\theta}|_{1,2,2}, \dots, \bar{\theta}|_{N_z,N_f,2}, \dots, \bar{\sigma}|_{N_z,N_f,2} \\ \bar{u}_r|_{1,2,L_m}, \bar{u}_r|_{2,2,L_m}, \dots, \bar{u}_r|_{N_z,2,L_m}, \bar{u}_r|_{1,3,L_m}, \dots, \bar{u}_r|_{N_z,N_f,L_m}, \bar{u}_z|_{1,2,L_m}, \dots, \bar{u}_z|_{N_z,N_f,L_m}, \bar{\theta}|_{1,2,L_m}, \dots, \bar{\theta}|_{N_z,N_f,L_m}, \dots, \bar{\sigma}|_{N_z,N_f,L_m} \end{bmatrix}^T \quad (25)$$

Finally, a Hankel transform inversion is performed to get the actual values of the variables in physical domain. Two categories of methods are adopted and results obtained by them are compared to guarantee the correctness of the solution in physical domain. The first method is carried out by the zero-point segmentation, and the scheme of the inversion is

$$f(r) \approx \mathbf{H}^{-1} \{ \bar{f}^m(\xi), [\xi_0, \xi_N] \} \approx \sum_{i=0}^{N-1} \left[v_i \sum_j \omega_j \bar{f}^m(\vartheta_{ij}) J_m(\vartheta_{ij} r) \vartheta_{ij} \right] \quad (26)$$

where $\mathbf{H}^{-1} \{ \bar{f}^m(\xi), [\xi_0, \xi_N] \}$ indicates performing a Hankel inversion for $\bar{f}^m(\xi)$ in $[\xi_0, \xi_N]$; $\vartheta_{ij} = (\xi_{i+1} + \xi_i)/2 + v_i \varpi_j$, $v_i = (\xi_{i+1} - \xi_i)/2$, in which $\xi_i (i = 0 \sim N)$ is the zero points of the m th Bessel function; $\xi_0 = 0$, N is set as large as 40 to make the calculation accurate; ω_j denote Gauss weighting coefficients and ϖ_j denote Gauss points. The second one is the MATLAB Toolbox for Infinite Integral of Products of Two Bessel Functions (IIPBF) proposed by Ratnanather et al. [33], and its accuracy has been demonstrated by recent studies [34–37]. More detailed scheme and codes for the IIPBF can refer to the study of Ratnanather et al. [33].

Verification

Mechanical loads

The behaviors of a porous medium under a uniform load are calculated and compared with those of Booker & Small [38] to verify the TDQM solution. The amplitude of the circular load is p_0 , and its radius $r_0 = h$, in which h is the thickness. The top of the medium is free for stress and totally permeable. The bottom is fixed and impermeable. Two cases with $\mu = 0$ and $\mu = 0.3$ are calculated. Please note that the temporal axes in the following figures are in logarithmic format such as Figs. 2, 4, 8 and so on. For convenience, the normalized displacement is defined as $u_z^* = Gu_z/r_0 p_0$, and the normalized time is $\tau = 2Gk_w t / \gamma_w h^2$.

From Fig. 2, we note that the vertical displacement of the medium increases over time and reaches a stable value at the center of the load action, which shows the consolidation process of the porous medium under external loads. The vertical displacement with $\mu = 0$ is significantly greater than that with $\mu = 0.3$. This is because the vertical stress will no longer cause horizontal deformation of the medium with $\mu = 0$, and the deformation will not be constrained in horizontal direction, leading to a significant increase in vertical displacement. The computational results by the TDQM are identical to the results obtained by Booker & Small [38], indicating the correctness of the proposed solution in calculating the displacements of homogeneous isotropic porous medium under external force loads.

Subsequently, to further verify the proposed solution for layered media, the pore pressure induced by circular loads on the surface of double-layered porous media is calculated. The thickness of each layer is $h/2$, and $\mu = 0.3$. The ratio of permeable coefficients $k_1/k_2 = 4$, and the ratio of shear moduli $G_1/G_2 = 1/4$. The calculated excess pore fluid For both displacement and pore pressure, as well as single-layered and multi-layered media, the solutions in this study are in good agreement with the results of existing literature, which demonstrates the correctness of the proposed theory.

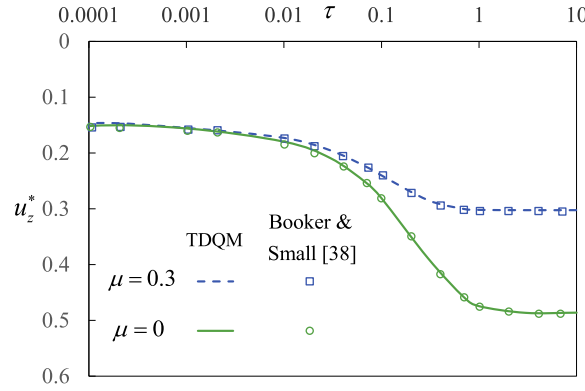


Fig. 2. The displacements induced by a uniform load at the original point.

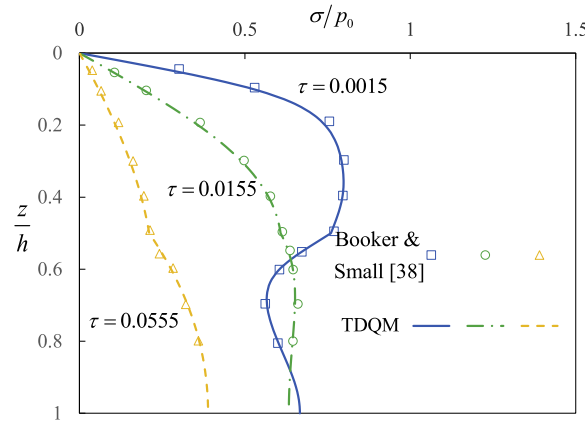


Fig. 3. The excess pore fluid pressure of double-layered media induced by a circular surface load.

Buried heat sources

To further discuss the applicability of the proposed theory to the coupled THM problem, we recalculate time-dependent behaviors of porous media with a deep buried point heat source. Results of the TDQM are compared with those by Savvddou & Booker [39]. A point heat source Q_0 is buried deep inside a half-space. Relevant thermal parameters include linear expansion coefficient α , thermal diffusion coefficient κ and thermal conductivity coefficient K . The consolidation factor is defined as $c_v = \frac{k_w}{\gamma_w}(\lambda + 2G)$. The Poisson's ratio μ is taken as 0.25 in this example. The observation point is located above the point heat source at a distance of R_0 . For convenience, we define the normalized temperature increment $\theta^* = 4\pi\theta KR_0/Q_0$ and normalized time $\tau = t\kappa/R_0^2$. From Fig. 4, results of the TDQM are consistent with the results of Savvddou and Booker [39], thus verifying the proposed TDQM solution. pressure along the centerline is shown in Fig. 3.

Numerical examples

Considering that some parameters of thermal-mechanical behaviors have been discussed in a recent study [32], we only investigate parameters of THM problems including α_s , α_f , k_w and n . Sensitivity analysis is the study of how the uncertainty in the output of a mathematical model or system (numerical or otherwise) can be influenced by its inputs. In this study, sensitivity analysis is carried out to quantitatively discuss the effects of parameters including α_s and α_f . Combining the perturbation method and differential method in sensitivity analysis, Liou & Yeh [40] defined a dimensionless sensitivity evaluation criterion for the k -th parameter P_k to the output result R ,

$$X_k = \frac{dR/R}{dP_k/P_k} = \frac{\partial R}{\partial P_k} \frac{P_k}{R} \quad (27)$$

where, $\partial R/\partial P_k$ represents the sensitivity of the variable P_k .

Eq. (27) can be approximated in a differential form as

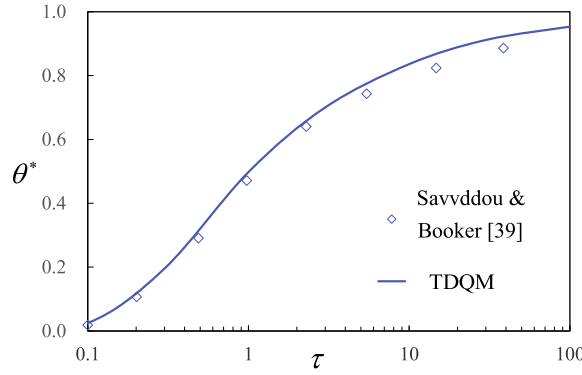


Fig. 4. The time-varying temperatures at observation points under deep buried heat source.

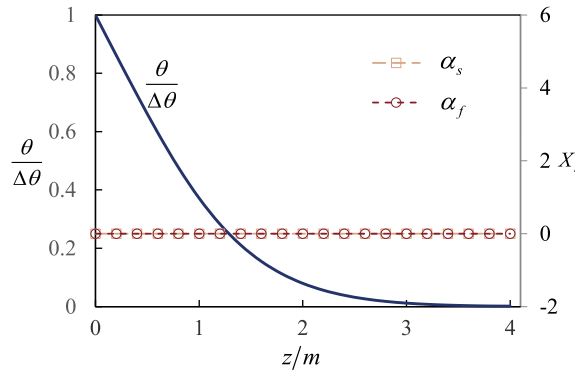


Fig. 5. The temperature increments along the centerline and relevant sensitivity parameters.

$$X_k \approx \frac{P_k}{R} \frac{\Delta R}{\Delta P_k} = \frac{P_k}{R(P_k)} \frac{R(P_k + \Delta P_k) - R(P_k)}{\Delta P_k} \quad (28)$$

Obviously, the smaller ΔP_k , the higher accuracy of the approximation. Referring to the research of Liou & Yeh [40], $\Delta P_k = 0.001P_k$ in this study.

The linear expansion coefficients

In this case study, we take the surface temperature increment $\Delta\theta = 10$ K, radius of temperature load $a = 2$ m, medium depth $h = 10$ m, $E = 25$ MPa, $G = 10$ MPa, $\mu = 0.25$, $K_\theta = 2$ W/(m · K), $k_w = 1$ E – 7 m/s, $\rho = 2000$ kg/m³, $C = 1200$ J/(kg · K), $n = 0.5$, $\alpha_s = 1$ E – 5/K, $\alpha_f = 7$ E – 5/K. The displacements, temperatures, and excess pore fluid pressure along the center line at 10 days are calculated, and α_s and α_f are analyzed by using the sensitivity parameter X_k .

The temperature increments along the centerline at 10 days and the sensitivity indicators of the parameters are shown in Fig. 5. The peak value of temperature appears near the temperature loading, and the temperature increment gradually returns to zero as the depth increases. The influence of α_s and α_f on the temperature results can be almost ignored, which means that if only the temperature response of the medium is required to be solved, the influence of α_s and α_f can be ignored in the coupled THM analysis.

Subsequently, the vertical displacements and sensitivity parameters along the centerline are calculated and displayed in Fig. 6. From Fig. 6, the maximum displacement occurs on the surface of the medium, where the uplift caused by the temperature is the most significant. For displacements, α_s and α_f have obvious effects, and their values of X_k are positive, that is, increasing α_s and α_f will lead to the increase of vertical displacements. This is because the linear expansion coefficient reflects the degree of thermal expansion of the solid skeleton and pore fluid. A larger linear expansion coefficient indicates that the thermal expansion of the medium at the same temperature is more significant, that is, the uplift displacement of the medium caused by heating increases. In comparison, the X_k value of α_s is much larger than that of α_f , which means that α_s has a more significant impact on the displacement, since the displacement is mainly determined by the expansion and contraction of the solid skeleton. As the depth increases, both X_k values of α_s and α_f remain stable, while their influences on displacements remains unchanged.

Then, we calculate the excess pore fluid pressure p and relevant sensitivity parameters along the centerline, and show them in Fig. 7. Since the surface of the medium is permeable, the excess pore fluid pressure at the surface is 0. p caused by surface temperature load gradually increases to its peak value along depth, and then gradually decreases. Both of α_s and α_f have significant effects on p , among which the effect of α_f is more significant.

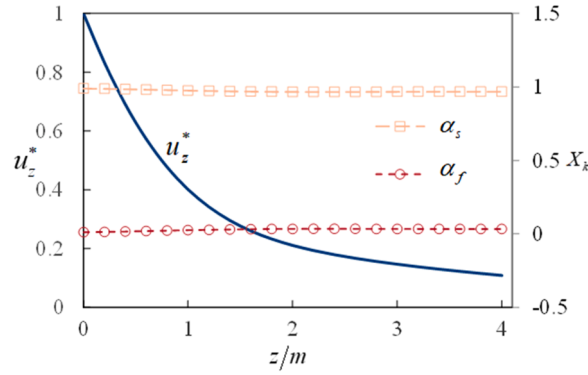


Fig. 6. The vertical displacements along the centerline and relevant sensitivity parameters.

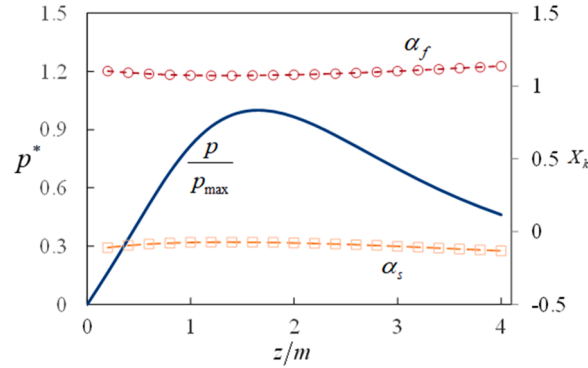


Fig. 7. The excess pore fluid pressure along the centerline and relevant sensitivity parameters.

The positive and negative signs of the X_k are opposite, which indicates they have opposite effects on p . From a mechanistic perspective, under the same temperature conditions, an increase in α_f will lead to more significant volume expansion of the fluid, while the medium skeleton limits the expansion of the fluid, resulting in an increase in excess pore pressure. On the contrary, an increase in α_s will make the solid skeleton expanding, resulting in an increase in pores and a decrease in excess pore fluid pressure.

From the above analysis, it can be concluded that in the THM problem, the influence of α_s and α_f can be ignored if we focus on the temperature increment response. However, both of α_s and α_f have an obvious impact on the displacement and p .

Permeable coefficient of fluid

To investigate the effect of permeable coefficient on the coupled THM response of media, a calculation example is designed as follows. The media is of three layers, and the load radius is one-fifth of the total thickness of the media. The parameters are shown in Table 1. The other parameters are set as the same in Section 4.1. In this example, $k_w = 1\text{E} - 8\text{m/s}$, and three cases with $0.1k_w$, k_w and $10k_w$ are calculated. For convenience, we define $u_z^* = u_z \times 10^5$ and $\tau = t/86400$, in which the unit of vertical displacement is m , and the unit of time t is s .

Fig. 8 shows the uplift displacements at the center of the load caused by temperatures. It can be seen that the final displacements of the medium under different permeable coefficients are the same, and the times of thermal consolidation are also the same (about 1000 days). The permeable coefficient mainly affects the displacement changes during thermal consolidation process. The fluid discharge is slower when the value of permeable coefficient is small, and the uplift displacement caused by temperature is more obvious, even greater than the final displacement. As the fluid gradually discharges, the displacement decreases and reaches a stable state. As the permeable coefficient increases, the pore fluid is more easily discharged, and the influence of the permeable coefficient on the thermal consolidation process gradually becomes less significant. Thus, in practical energy geotechnics, it is necessary to determine the permeable coefficient to accurately predict the uplift of the media under temperature over time.

Figs. 9 and 10 show the changes in p along depth with $0.1k_w$ and k_w . Obviously, fluid permeability has a significant impact on the peak value of p . With a surface temperature load of 10°C , p inside the medium can reach around 1.5 kPa . The location where the peak value appears becomes deeper with time, while the peak values become smaller. When the permeable coefficient of the fluid increases, the peak value of p will significantly decrease, which is caused by the easier discharge of pore fluid. Thus, in practical engineering, the drainage conditions of the media can be improved by pre-burying drainage channels to prevent the internal pore pressure increasing to a critical value, leading to an engineering damage.

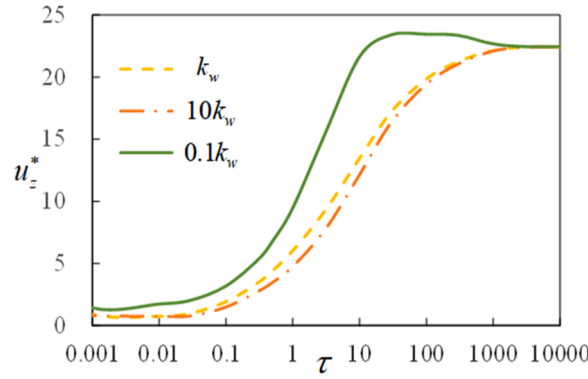


Fig. 8. The vertical displacements at the origin point with time.

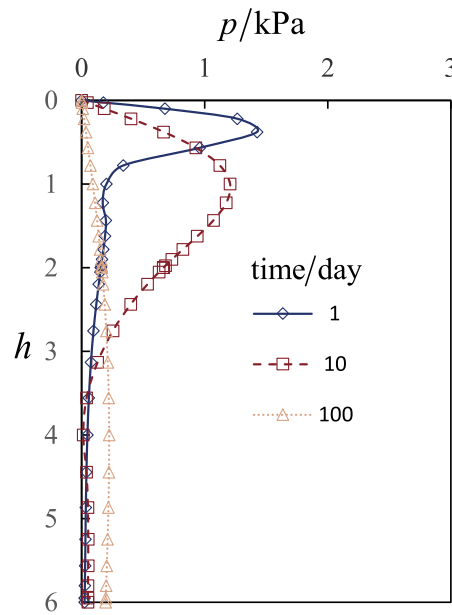


Fig. 9. Excess pore fluid pressure with $0.1k_w$.

Porosity

In practical engineering, the porosity of a medium often varies within a certain range. To explore the influence of porosity on the THM behaviors of porous media, we design a calculation example as follows: three cases with porosity of 0.3, 0.5, and 0.7; the fluid permeable coefficient $k_w = 1 \times 10^{-7}$ m/s, and the other parameters are the same as the example in Section 4.2.

As shown in Fig. 11, the uplift displacement caused by heating on the surface is independent of n , and the final uplift displacements and thermal consolidation process are almost the same with different porosities. Therefore, in the engineering of geothermal exchangers, the influence of porosity can be ignored when exploring the displacements caused by temperature or heat sources.

From Figs. 12–14, the influence of porosity on the internal excess pore fluid pressure is significant, and when the porosity is 0.7, the overall excess pore pressure inside the medium is greater than the other two cases.

This is because for the porous medium, a larger porosity means that the pore fluid accounts for a larger proportion of the total volume of the medium. Correspondingly, the volume change caused by thermal expansion of the pore fluid is more significant, resulting in a higher p when the fluid is not discharged in time. In addition, p inside the medium will gradually decrease over time, and eventually dissipate towards zero.

Conclusions

A novel TDQM solution is derived to investigate the coupled THM problem in layered isotropic porous media, and constant temperature increment loadings are considered in this study. By virtue of the integral transform and the discretization rule, the spatial

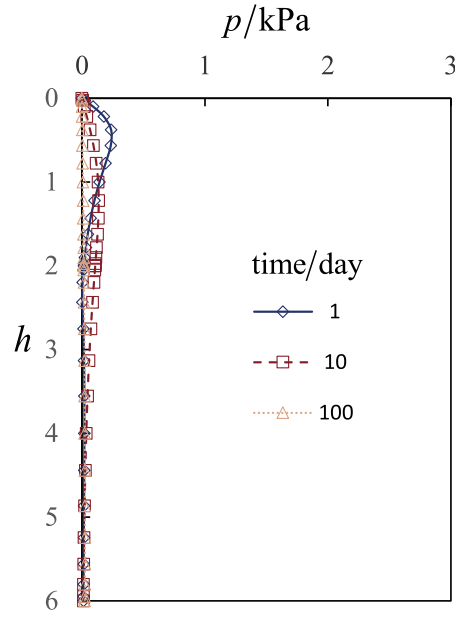


Fig. 10. Excess pore fluid pressure with k_w .

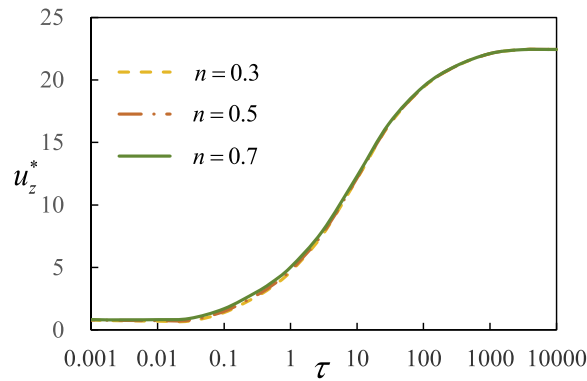


Fig. 11. The vertical displacements at the origin point with time.

domain along depth are discretized. The TDQM solution is verified through recalculating examples of existed literatures. Also, sensitivity analyses are carried out and the linear expansion coefficients of solid skeleton and fluid, the permeable coefficient and porosity are discussed. The main conclusions are drawn as follows:

(1) α_s and α_f both have an obvious impact on the excess pore fluid pressure, and the influence of α_f is more significant. While α_s only has a significant impact on the vertical displacements.

(2) The final displacements and thermal consolidation times of the medium under different permeable coefficients are the same, and the permeable coefficient mainly affects the change process of displacement and pore pressure. When the permeable coefficient of the fluid increases, the peak value of excess pore pressure will significantly decrease, and the peak value location becomes deeper with time.

(3) In the engineering of geothermal exchangers, the influence of porosity can be ignored when discussing the displacements caused by temperature or heat sources. The larger the porosity of the medium, the more significant the excess pore pressure caused by temperature inside the medium.

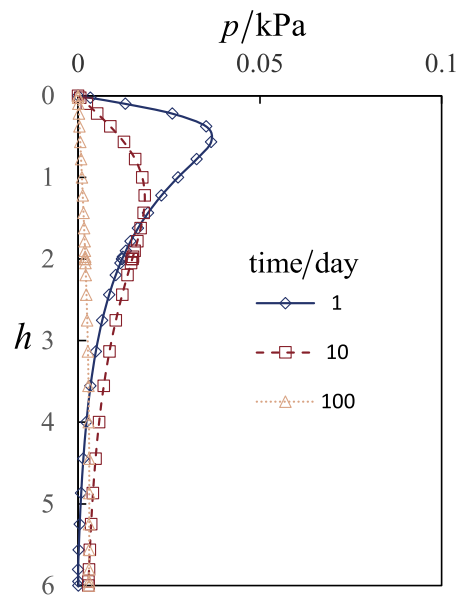


Fig. 12. Excess pore fluid pressure with $n = 0.7$.

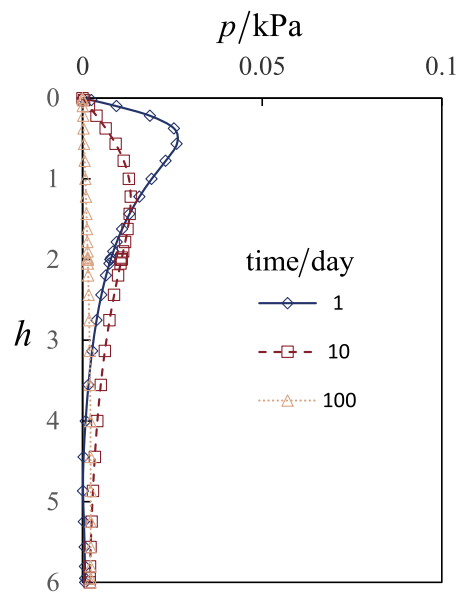


Fig. 13. Excess pore fluid pressure with $n = 0.5$.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

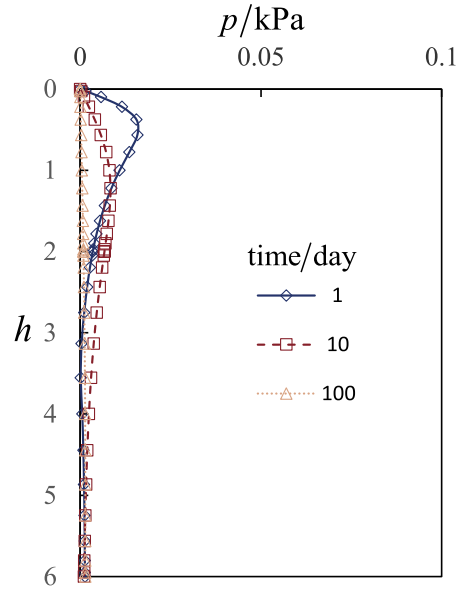


Fig. 14. Excess pore fluid pressure with $n = 0.3$.

Acknowledgements

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Data availability statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Appendix A

The coefficient matrices are

$$\mathbf{A} = \begin{bmatrix} A_{11} & A_{12} & \cdots & A_{1N_z} \\ A_{21} & A_{22} & & \\ \vdots & & \ddots & \vdots \\ A_{N_z 1} & \cdots & & A_{N_z N_z} \end{bmatrix}$$

$$\mathbf{A}^2 = \begin{bmatrix} A_{11}^2 & A_{12}^2 & \cdots & A_{1N_z}^2 \\ A_{21}^2 & A_{22}^2 & & \\ \vdots & & \ddots & \vdots \\ A_{N_z 1}^2 & \cdots & & A_{N_z N_z}^2 \end{bmatrix}$$

$$\hat{\mathbf{A}} = \text{diag}(\mathbf{A}, \dots, \mathbf{A})_{N_t-1}$$

$$\hat{\mathbf{A}}^2 = \text{diag}(\mathbf{A}^2, \dots, \mathbf{A}^2)_{N_t-1}$$

$$\mathbf{B}_{ij} = \text{diag}(B_{ij}, \dots, B_{ij})_{N_z}$$

$$\hat{\mathbf{B}} = \begin{bmatrix} \mathbf{B}_{22} & \mathbf{B}_{23} & \cdots & \mathbf{B}_{2N_t} \\ \mathbf{B}_{32} & \mathbf{B}_{33} & & \\ \vdots & & \ddots & \vdots \\ \mathbf{B}_{N_t 2} & \cdots & & \mathbf{B}_{N_t N_t} \end{bmatrix}$$

$$\hat{\mathbf{B}} = [\mathbf{B}_{21} \quad \mathbf{B}_{31} \quad \cdots \quad \mathbf{B}_{N_t 1}]^T$$

$$\hat{\mathbf{C}} = \begin{bmatrix} B_{22}\mathbf{A} & B_{23}\mathbf{A} & \cdots & B_{2N_f}\mathbf{A} \\ B_{32}\mathbf{A} & B_{33}\mathbf{A} & & \vdots \\ \vdots & & \ddots & \\ B_{N_f2}\mathbf{A} & \cdots & & B_{N_fN_f}\mathbf{A} \end{bmatrix}$$

$$\hat{\mathbf{C}} = [B_{21}\mathbf{A} \quad B_{31}\mathbf{A} \quad \cdots \quad B_{N_f1}\mathbf{A}]^T$$

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